

Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.16846	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	75.34422	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1443.8326	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	0.0	3.0	n/a	295.04778	2	0	0	0.66215	16.925
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	0.0	3.0	n/a	255.20844	1	0	0	0.935765	12.91
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.273435	0.273435	141.46268	0	0	0	0.0	0.0

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.16846.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.

- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 75.34422.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1443.8326.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]

- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 295.04778.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
 - X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 255.20844.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
 - X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 141.46268.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary (Conservative)

Aspect	Observation
Direct Synthesis	direct_generate_plus_one_repair failed fully on feasibility (0% feasible) with high penalty gap (3.0).
Budget-Matched Independent Search	budget_matched_no_replay also had 0% heldout feasibility and same penalty gap (3.0), indicating failure.
Replay-Aware Iterative Search	replay_solver_evolution had full heldout feasibility (100%) and a feasible gap of 0.273435, but no improvement over nearest neighbor baseline.
Fixed-Baseline Context	Best baseline: phase9_closeout_baseline_clarke_wright_savings had 100% feasibility, lowest feasible gap (0.073766), runtime 75ms, and modest robustness dispersion (0.0078).
Feasibility	All baseline methods had 100% feasibility; learned/iterative methods without replay failed feasibility on heldout.
Objective Gap	Clarke-Wright savings best gap (0.0738), followed by nearest neighbor (0.2734), replay solver equals nearest neighbor (0.2734). Regret insertion had worst gap (0.4664). Learned methods without replay gap 3.0 (failure).
Runtime	Regret insertion slowest (~1444 ms), Clarke-Wright moderate (~75 ms), replay solver longer than Clarke-Wright (~141 ms), nearest neighbor fastest (~42 ms). Learned methods slower than baselines (~255-295 ms).

Aspect	Observation
Robustness	Baselines showed low dispersion (best: nearest neighbor 0.0028, replay solver similar dispersion), regret insertion most dispersed (0.0869). Learned methods had zero dispersion reported but infeasible.
Learned Condition vs Fixed Baselines	No learned method improved over fixed baselines; direct synthesis and budget-matched search failed feasibility completely.
Replay vs No-Replay Control	Replay-aware method (replay_solver_evolution) had 100% feasibility and feasible gap 0.2734, outperforming no-replay (budget_matched_no_replay), which had 0% feasibility and gap 3.0. Replay clearly better than no replay here.

Key Conclusions

- ****Clarke-Wright savings baseline remains best overall**** on heldout: highest feasibility, lowest objective gap, moderate runtime.
- ****Replay-aware iterative search maintains full feasibility**** and matches nearest neighbor in gap but is slower; ****better than budget-matched no replay control, which failed feasibility.****
- ****Direct synthesis and budget-matched independent search approaches failed to produce feasible solutions.****
- ****No learned condition clearly outperformed fixed baselines on feasibility or objective gap.****
- Replay mechanism improves robustness and feasibility compared to no replay in budget-matched setting.